Thermodynamics

Grau en Enginyeria Matemàtica i Física

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Second Quadrimester 2024

4. Formal structure. Thermodynamic potentials

4.1. Formal Structure

Postulate II

The existence of the state variable S is of great relevance for Thermodynamics, as it gives rise to the discovery of **the internal mathematical structure of the theory**, as that of Newton's laws in Mechanics. Up to this point, what we have learned is a collection of experimental and derived facts, much like the botanists who in the 19th Century were collecting plants and animals, to catalog them.

4.1. Formal Structure

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4.1. Formal Structure

Postulate II

In order to introduce this formal structure, as a starting point we must consider the First Law of thermodynamics, which we will write differentially as

$$dU = dQ + dW$$

If we further consider that this variation in U is due to a **reversible** change, we will have that the work will be written as

$$dW = -pdV$$

On the other hand, according to what we saw earlier, we can also write

$$dQ = TdS$$

Then, putting these results together, we arrive at the so-called **Gibbs equation** (not to be confused with the *Gibbs function*, which we will see later),

$$\boxed{dU = TdS - PdV} \qquad (1)$$

4.1. Formal Structure

Postulate II

Consequences of Gibbs equation

- In view of the form of the Gibbs equation, we can say that *the internal energy function U is a natural function of the variables S and V*. It is also a function of the number of moles, but we will talk about this when we introduce the exchange of matter. For now, we consider the number of moles to be constant.

$$\boxed{U = U(S, V)} \tag{2}$$

This equation is what is called the **fundamental equation**. The fundamental equation **contains all the thermodynamic information about a system.**

4.1. Formal Structure

Postulate II

Consequences of Gibbs equation

- As a mathematical function, after identifying the independent variables, we can write

$$dU = \left.\frac{\partial U}{\partial S}\right|_V dS + \left.\frac{\partial U}{\partial V}\right|_S dV$$

By comparing this equation to the Gibbs equation,

$$dU = TdS - PdV$$

we can conclude that the **equations of state represent first derivatives of the fundamental equation:**

$$\boxed{T = \left.\frac{\partial U}{\partial S}\right|_V ; \quad P = -\left.\frac{\partial U}{\partial V}\right|_S} \tag{3}$$

4.1. Formal Structure

Postulate II

Consequences of the Gibbs equation: Maxwell relations and thermodynamic consistency

- Based on the fact that

$$dU = \left.\frac{\partial U}{\partial S}\right|_V dS + \left.\frac{\partial U}{\partial V}\right|_S dV = TdS - PdV$$

we can establish a relation, called **Maxwell relation**

$$\boxed{\frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S} \Rightarrow \left.\frac{\partial T}{\partial V}\right|_S = -\left.\frac{\partial P}{\partial S}\right|_V} \quad (4)$$

4.1. Formal Structure

Postulate II

Let's check that the equations of state of the ideal gas are **consistent**. To do this, we better rewrite the Gibbs equation in its *entropic form*, which says $S = S(U, V)$ (where U and V are the independent variables)

$$dS = \frac{1}{T}dU + \frac{P}{T}dV$$

From the previous relation we can write

$$\frac{\partial^2 S}{\partial U \partial V} = \frac{\partial^2 S}{\partial V \partial U} \Rightarrow \left.\frac{\partial 1/T}{\partial V}\right|_U = \left.\frac{\partial P/T}{\partial U}\right|_V$$

Then, we rewrite the ideal (monatomic) gas equations of state $PV = NRT$ and $U = \frac{3}{2}NRT$ as

$$\frac{P}{T} = \frac{NR}{V}; \quad \frac{1}{T} = \frac{3NR}{2U}$$

We can verify that, indeed,

$$\left.\frac{\partial 1/T}{\partial V}\right|_U = \left.\frac{\partial P/T}{\partial U}\right|_V = 0$$

In fact, with this knowledge, and knowing that $\left.\frac{\partial P/T}{\partial U}\right|_V = 0$, **we could have inferred that the internal energy of an ideal gas depends only on its temperature** and, therefore, we could have deduced the result of Joule's experiment without ever doing it:

$$-\frac{1}{T^2}\left.\frac{\partial T}{\partial V}\right|_U = 0 \quad \Rightarrow \quad \left.\frac{\partial U}{\partial V}\right|_T = -\left.\frac{\partial U}{\partial T}\right|_V \left.\frac{\partial T}{\partial V}\right|_U = 0 \;\; \text{q.e.d.}$$

4.1. Formal Structure

Postulate II

Extension to include matter exchange with the surroundings

Gibbs realized that the exchange of matter should be included in the **Gibbs equation**, eq. (1). In other words, to incorporate the change in internal energy when the moles of a given substance are varied. Then, by analogy with the work expression $-pdV$, which is the product of an extensive variable times its intensive conjugate, or that of heat, TdS, Gibbs proposed the **chemical potential** μ as the intensive variable which is the conjugate of the number of moles, and wrote its contribution to the internal energy as μdN.

$$\boxed{dU = TdS - PdV + \mu dN} \tag{5}$$

This indicates that there are three equations of state for a 1-component system, eq. (3) plus

$$\mu = \frac{\partial U}{\partial N}|_{S,V}$$

4.1. Formal Structure

Postulate II

Extension to include matter exchange with the surroundings: Maxwell relations

After extending the Gibbs equation with the chemical potential, eq. (5) (for a monocomponent system: $dU = TdS - PdV + \mu dN$), we have two Maxwell relations in addition to eq. (4)

$$\left.\frac{\partial T}{\partial N}\right|_{S,V} = \left.\frac{\partial \mu}{\partial S}\right|_{V,N}$$
$$-\left.\frac{\partial P}{\partial N}\right|_{S,V} = \left.\frac{\partial \mu}{\partial V}\right|_{S,N} \qquad (6)$$

4.1. Formal Structure

Postulate II

Extension to include matter exchange with the surroundings

For a system with n components (species), $U = U(S, V, N_1, \ldots, N_n)$ and the final form of the Gibbs equation becomes

$$dU = TdS - PdV + \sum_{i=1}^{n} \mu_i dN_i$$

which includes the chemical potentials of all the components, μ_i. This is the **central equation of thermodynamics**.

Now there are $2 + n$ equations of state,

$$T = \left.\frac{\partial U}{\partial S}\right|_{V,N_1,\ldots,N_n} ; \quad P = -\left.\frac{\partial U}{\partial V}\right|_{S,N_1,\ldots,N_n} ; \quad \mu_i = \left.\frac{\partial U}{\partial N_i}\right|_{S,V,N_{j\neq i}} (i = 1, \ldots, n)$$

All these are related to one another by the conditions of thermodynamic consistency, like eqs.(4) and (6).

4.1. Formal Structure

Postulate II

Postulate II

There exists a function U of the extensive parameters of any multi-component system $S, V, N_1, \ldots N_n$, defined at all equilibrium states, and which has the following property: the values taken by the extensive parameters of a thermally isolated system, connected to a source of reversible work, in the absence of internal contraints, are those that minimize U.

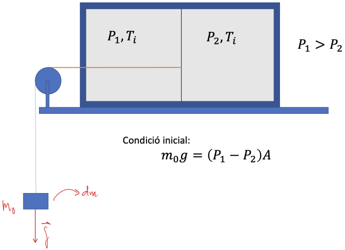
This postulate allows an equivalent **entropic formulation** (see [2]):

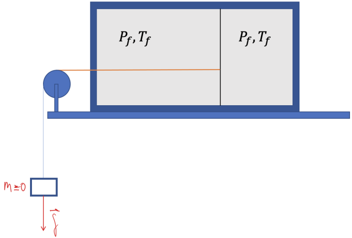
There exists a function (called entropy S) of the extensive parameters of any compound system, defined at all equilibrium states, and which has the following property: the values taken by the extensive parameters of an isolated system, in the absence of internal constraints, are those that maximize entropy.

4.1. Formal Structure

Postulate II

Consider this thermally-isolated composite system connected to a reversible work source:





The initial energy is U_i and the final energy is U_f. After liberating the constraint (not shown) the system will evolve reversibly without heat exchange (thus at constant S). We have that $U_f - U_i = W < 0$. And if we perturbed the system from its final state, then: $\delta U = U - U_f = \delta W > 0$. Ergo, the final equilibrium state coincides with the minimum of U. This is the solution to the fundamental problem of thermodynamics.

4.1. Formal Structure

Postulate II

Extensiveness-related properties of U: Euler form

The fact that U is written in terms of **extensive** functions has important formal implications. Upon multiplying the size of the system by an arbitrary factor λ:

$$U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$$

Deriving both members with respect to λ, we can write

$$\left.\frac{\partial U}{\partial(\lambda S)}\right|_{V,N} \frac{\partial(\lambda S)}{\partial \lambda} + \left.\frac{\partial U}{\partial(\lambda V)}\right|_{S,N} \frac{\partial(\lambda V)}{\partial \lambda} + \left.\frac{\partial U}{\partial(\lambda N)}\right|_{S,V} \frac{\partial(\lambda N)}{\partial \lambda} = U(S, V, N)$$

The above is valid for any λ in particular for $\lambda = 1$. Therefore, by using equation (3), we arrive at the **Euler form** of the fundamental equation:

$$\boxed{U(S, V, N) = TS - PV + \mu N} \tag{7}$$

Specifically, this means: $U(S, V, N) = T(S, V, N)\, S - P(S, V, N)\, V + \mu(S, V, N)\, N$.

4.1. Formal Structure

Postulate II

Extensiveness-related properties of U: Gibbs-Duhem equation

The Gibbs-Duhem equation arises from the fact that **not all intensive variables are independent**. Indeed, if we differentiate U in its Euler form (7), we obtain

$$dU = TdS + SdT - PdV - VdP + \mu dN + Nd\mu$$

And if we compare this to the Gibbs equation (5) ($dU = TdS - PdV + \mu dN$), the following relationship between the intensive variables must be satisfied:

$$\boxed{0 = SdT - VdP + Nd\mu} \tag{8}$$

known as the **Gibbs-Duhem equation**. To integrate this equation one would need the equations of state, to be able to express the extensive variables as functions of the intensive ones. If we are missing one eq. of state, we can integrate GD to obtain it, so we can re-compose the fundamental equation except for an unknown constant.

4.1. Formal Structure

Postulate II

Chemical potential of an ideal gas

By considering the Gibbs-Duhem equation, we see that the chemical potential of an ideal gas cannot just be anything, but must be linked to the other equations of state. Indeed, if we consider isothermal conditions, we can write,

$$0 = -VdP + Nd\mu$$

After introducing the equation of state, we arrive at

$$d\mu = \frac{RT}{P} dP$$

wherefrom we get the third and final equation of state for the ideal gas

$$\mu(T, P) = \mu(T, P^\circ) + RT \ln \frac{P}{P^\circ}$$

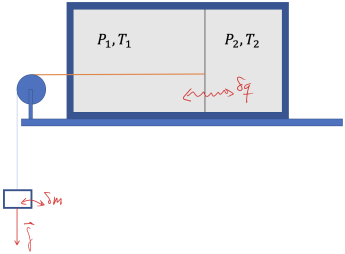
If we take $P^\circ = 1$ bar, we have $\mu(T, P^\circ) \equiv \mu^\circ(T)$ or the standard chemical potential at temperature T.

4.1. Formal Structure

Postulate II

Equilibrium and stability conditions

We can now show that the postulate of the minimum of U under conditions of $S = \text{ct}$ and $V = \text{ct}$ allows us to generalize the equilibrium conditions obtained by mechanical arguments. In addition, we can also establish facts about stability.



Let us assume that the system shown is in equilibrium, and that this occurs at the minimum of $U(V, S)$ under these conditions. If we disturb the equilibrium state with reversible work δW and reversible heat δQ, with the contraints $V_1 + V_2 = V$ and $S_1 + S_2 = S$, with constant V and S, we can write,

$$\begin{aligned}\delta U =&\delta(U_1 + U_2) = -(P_1 - P_2)\delta V_1 + (T_1 - T_2)\delta S_1 + \\ &+\frac{1}{2}\left(-\left.\frac{\partial P_1}{\partial V_1}\right|_s (\delta V_1)^2 + 2\left.\frac{\partial T_1}{\partial V_1}\right|_s \delta V_1 \delta S_1 + \left.\frac{\partial T_1}{\partial S_1}\right|_v (\delta S_1)^2\right) + (1 \longleftrightarrow 2) + \ldots > 0\end{aligned}$$

4.1. Formal Structure

Postulate II

Equilibrium and stability conditions

Therefore, since δV_1 and δS_1 are arbitrary and have a sign, the linear terms must be null at equilibrium. This enforces that

$$P_1 = P_2 \quad , \quad T_1 = T_2$$

These are the conditions which we have determined *a priori* as facts of experience. The first one comes from mechanics and the second is what has allowed us to define temperature.

In addition, the second-order terms must be positive (for U to be a minimum). At this point, we will first introduce some common coefficients to help us in the analysis:

$$\kappa_s \equiv -\frac{1}{V}\left.\frac{\partial V}{\partial P}\right|_s \quad \text{Adiabatic compressibility} \tag{9}$$

$$C_V \equiv T \left.\frac{\partial S}{\partial T}\right|_V \quad \text{Heat capacity at constant volume} \tag{10}$$

4.1. Formal Structure

Postulate II

Equilibrium and stability conditions

$$\kappa_T \equiv -\frac{1}{V}\left.\frac{\partial V}{\partial P}\right|_T \quad \text{Isothermal compressibility} \tag{11}$$

$$\alpha \equiv \frac{1}{V}\left.\frac{\partial V}{\partial T}\right|_P \quad \text{Thermal expansion coefficient} \tag{12}$$

The term $\left.\frac{\partial T_1}{\partial V_1}\right|_s$ can be expressed in terms of these coefficients

$$\left.\frac{\partial T_1}{\partial V_1}\right|_s = -\frac{T\alpha}{C_V\kappa_T} \tag{13}$$

4.1. Formal Structure

Postulate II

Equilibrium and stability conditions

According to this, the following conditions must be satisfied for the system to be in thermodynamic equilibrium [6]:

$$\frac{1}{V\kappa_s} > 0 \tag{14}$$

$$\frac{T}{C_V} > 0 \tag{15}$$

$$\begin{vmatrix} \frac{1}{V\kappa_s} & -\frac{T\alpha}{C_V\kappa_T} \\ -\frac{T\alpha}{C_V\kappa_T} & \frac{T}{C_V} \end{vmatrix} > 0 \tag{16}$$

From here we deduce that

$$\kappa_s > 0; \quad C_V > 0; \quad \text{i} \quad \frac{1}{\kappa_s} > \frac{T\alpha^2 V}{C_V \kappa_T^2} \tag{17}$$

4.2. Themodynamic Potentials

Enthalpy revisited

We have seen that the existence of entropy and internal energy has led us to state that the function $U = U(S, V, N)$ contains all the thermodynamic information, and that the Gibbs equation,

$$dU = TdS - PdV + \mu dN$$

is the fundamental equation.

From this perspective, we can now revisit enthalpy. Recall that $H \equiv U + PV$. This transformation is a so-called **Legendre transformation**. According to this, we will have

$$dH = dU + PdV + VdP = TdS - PdV + \mu dN + PdV + VdP$$

Therefore,

$$\boxed{dH = TdS + VdP + \mu dN \quad \Rightarrow \quad H = H(S, P, N)}$$

From this expression we can see that the enthalpy is a function of the natural variables (S, P, N), that is, we have replaced the volume with **its conjugate variable**, the pressure, as independent variable.

In what follows, we will apply Legendre transformations to obtain a powerful tool for solving thermodynamic problems.

4.2. Themodynamic Potentials

Enthalpy revisited

Useful work and minimum condition

Enthalpy will be useful for dealing with systems at constant S and P.[2] First, consider a *closed* composite system in mechanical equilibrium with a *source of volume* (system that exchanges volume at constant external pressure, P_{ext}). In this case, for the 'total system' (system + volume source) in equilibrium with $P = P_{ext}$:

$$0 = dU^{tot} = d(U + U^{VS}) = dU - P_{ext} dV^{VS} = dU + P_{ext} dV = dU + PdV =$$

$$dH|_{P=P_{ext}} = d(U + PV) = 0$$

Therefore, **the equilibrium value of any unconstrained internal parameter of a system which is in equilibrium with a volume source, minimizes the system's enthalpy at constant P equal the source pressure**.

Now, consider the same closed system and volume source, but the former is connected to a *source of reversible work*. In this case, the First Law dictates that $dW^{useful} = -dU^{tot}$, therefore,

$$\boxed{\Delta H|_{P=P_{ext}} = -W^{useful}}$$

4.2. Themodynamic Potentials

Enthalpy revisited

Joule-Thomson Coefficient

In this process, a certain number of moles dN pass through a porous plug from a compartment (left side in figure) at a pressure P_1, to a compartment (right side in figure) at a lower pressure P_2. The system, as a whole, is thermally isolated.

Due to the conservation of energy, the change in the internal energy of the whole varies according to

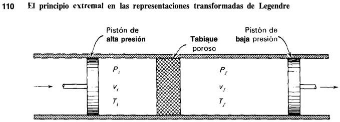


Fig. 6.2 Representación esquemática del proceso Joule-Thomson.

$$dU = dU_1 + dU_2 = dW_1 + dW_2 = -P_1 dV_1 - P_2 dV_2$$

The link between the variations is the number of moles transferred, dN, that leave the left side and go to the right. If we introduce molar variables (u is the energy per mole and v is the molar volume), we will state

$$dU_1 + P_1 dV_1 = (u_1 + P_1 v_1)(-dN) \text{ and } dU_2 + P_2 dV_2 = (u_2 + P_2 v_2) dN$$

The expressions in parentheses are the molar *enthalpies* on both sides of the porous plug; so,

$$-h_1 dN + h_2 dN = 0 \quad \Rightarrow \quad h_1 = h_2$$

Therefore, the Joule-Thomson process is a process at **constant enthalpy**.

If we are interested in the temperature variation as a function of the pressure variation across the plug, we will seek the **Joule-Thompson coefficient**, μ_{JT}

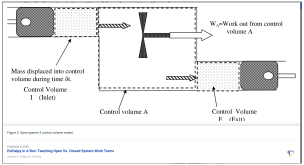
$$\boxed{\mu_{JT} \equiv \left.\frac{\partial T}{\partial P}\right|_H}$$

4.2. Themodynamic Potentials

Enthalpy revisited

Open systems

This result can be generalized to systems in steady flow (very common in chemical and mechanical engineering) [7].



The useful work (AKA *shaft work*) that we can extract from it coincides with the enthalpy change, that is to say, (per unit time)

$$\boxed{\Delta \dot{H} = -\dot{W}^{useful}}$$

if the system is thermally isolated, and viscous losses can be neglected.

More generally, H is the *stagnation enthalpy*, which accounts for the kinetic energy of the fluid: $\dot{H} = \dot{m}H \equiv \dot{m}(h + \frac{\nu^2}{2})$, where $\dot{m}$ is the mass rate through the system (constant in steady state), h is the enthalpy per unit mass, and ν is the average speed of the fluid. h is equal to $u + P/\rho$, where u is internal energy per unit mass, P is pressure, and ρ is density. If the fluid is incompressible (a liquid), and the inlet and outlet are equally sized (equal cross sectional areas), the kinetic term does not change and $\dot{m}\Delta h = -\dot{W}^{useful}$.

4.2. Themodynamic Potentials

Helmholtz free energy

We now consider a Legendre transformation in which we want to replace entropy with temperature (conjugate variables). The thermodynamic potential is called **Helmholtz free energy**, which we will represent as F (*free energy*) although A (*arbeit*, in German is also often used).

We define the Helmholtz free energy [3] as

$$F \equiv U - TS$$

To find the natural variables of F we must resort to the Gibbs equation, together with the previous definition, that is to say,

$$dF = dU - TdS - SdT = TdS - PdV + \mu dN - TdS - SdT$$

wherefrom we get

$$\boxed{dF = -SdT - PdV + \mu dN \quad \Rightarrow \quad F = F(T, V, N)} \qquad (18)$$

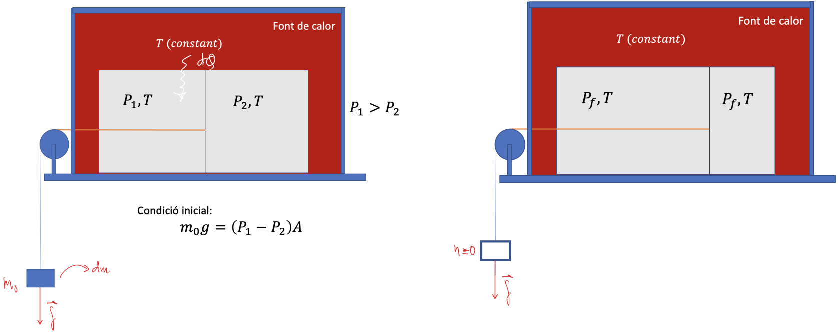
Therefore, the natural variables of F are T, V, N.

4.2. Thermodynamic Potentials

Helmholtz free energy

Useful work and minimum condition

The Helmholtz free energy will be useful for dealing with systems at constant T and V. To see this, consider again the closed (composite) system that does reversible work, in contact with a heat source (system that exchanges heat but does not change its temperature), that is,



4.2. Themodynamic Potentials

Helmholtz free energy

Useful work and minimum condition

We consider the variation of the internal energy of the composite system only, taking into account that the heat has been transferred from the heat source reversibly:

$$\Delta U = W + Q \quad \text{with} \quad Q = T\Delta S$$

$$\Delta U - T\Delta S = W$$

We see that, **since the temperature is constant**, $\Delta F = \Delta U - \Delta(TS) = \Delta U - T\Delta S$, and

$$\boxed{\Delta F = W}$$

In other words, **when the system is isothermal and the volume is constant, the reversible work we can extract from it is equal to the variation of the Helmholtz free energy**. Hence the names *accessible work*, *useful work*, and *free energy*.

The condition $\Delta F = W$ indicates that the system will be able to do work until reaching the minimum in its free energy F. We can say, then, that the system will reach the final equilibrium **when F will reach its minimum**.

4.2. Themodynamic Potentials

Helmholtz free energy

Useful work and irreversibility

If the process were irreversible, then we would have the Clausius inequality (see Topic 3)$\Delta S > \frac{Q^{irr}}{T}$ (where T is the heat source temp.). Then, $Q^{irr} < T\Delta S$. Now, by using the First Law and this inequality, we can write

$$\Delta U - Q^{irr} = W^{irr} \Rightarrow \Delta F = \Delta U - T\Delta S < W^{irr}$$

Since the work on the system is negative ($W^{irr} < 0$), we conclude that the obtained work ($-W^{irr}$) is less than the absolute change in F, that is,

$$\boxed{\Delta F < W^{irr}} \text{ or } |\Delta F| > |W^{irr}|$$

In irreversible processes, **not all the free energy is converted into work!**

Minimum principle at T and V constant: $\Delta F < W^{irr}$

In any process that occurs spontaneously in a system at constant T and V, the Helmhotz free energy will eventually decrease to its minimum.

4.2. Themodynamic Potentials

Helmholtz free energy

Other properties

If we know $F(T, V, N)$, we can also deduce the state equations. As before, these arise from comparing the differential dF with equation (18). By comparing term by term we deduce that

$$S = -\left.\frac{\partial F}{\partial T}\right|_{V,N}; \quad P = -\left.\frac{\partial F}{\partial V}\right|_{T,N}; \quad \mu = \left.\frac{\partial F}{\partial N}\right|_{T,V} \tag{19}$$

In the same way, we obtain three relations of thermodynamic consistency (Maxwell's relations), which now involve different variables than were obtained with the internal energy:

$$\left.\frac{\partial S}{\partial V}\right|_{T,N} = \left.\frac{\partial P}{\partial T}\right|_{V,N}; \quad -\left.\frac{\partial S}{\partial N}\right|_{T,V} = \left.\frac{\partial \mu}{\partial T}\right|_{V,N}; \quad -\left.\frac{\partial P}{\partial N}\right|_{T,V} = \left.\frac{\partial \mu}{\partial V}\right|_{T,N}$$

4.2. Themodynamic Potentials

Helmholtz free energy

The ideal gas revisited

We can calculate the Helmholtz free energy of the ideal gas using the information we have found so far. We have

$$U(T,N) - U(T_0,N) = \int_{T_0}^{T} dT'\, C_V(T',N)$$

$$S(T,V,N) - S(T_0,V_0,N) = \int_{T_0}^{T} dT'\, \frac{C_V(T',N)}{T'} + NR\ln\frac{V}{V_0}$$

Then, since $F = U - TS$

$$F(T,V,N) = U(T_0,N) + \int_{T_0}^{T} dT'\, C_V(T',N) - T\left(S(T_0,V_0,N) + \int_{T_0}^{T} dT'\, \frac{C_V(T',N)}{T'} + NR\ln\frac{V}{V_0}\right)$$

By defining $F(T_0,V_0,N) = U(T_0,N) - T_0 S(T_0,V_0,N)$ we arrive at the functional form of the fundamental equation

$$\boxed{F(T,V,N) = F_0 - (T-T_0)S_0 + \int_{T_0}^{T} dT'\, C_V(T',N) - T\left(\int_{T_0}^{T} dT'\, \frac{C_V(T',N)}{T'} + NR\ln\frac{V}{V_0}\right)} \quad (20)$$

where we wrote $F_0 \equiv F(T_0,V_0,N)$ and $S_0 = S(T_0,V_0,N)$. The interested reader can now derive Boyle's law from this equation, by applying the corresponding expression among Eqs. (19).

4.2. Themodynamic Potentials

Gibbs free energy

Let us now consider a Legendre transformation in which we want to replace entropy by temperature and volume by pressure (pairs of conjugate variables). The thermodynamic potential is called **Gibbs free energy** (or Gibbs function) [4], which we will represent as G, and define as

$$G \equiv U - TS + PV$$

To find the natural variables of G we must resort to the Gibbs equation (not to be confused with the Gibbs function), along with the previous definition, that is,

$$dG = dU - TdS - SdT + PdV + VdP = TdS - PdV + \mu dN - TdS - SdT + PdV + VdP$$

wherefrom we get that

$$\boxed{dG = -SdT + VdP + \mu dN \quad \Rightarrow \quad G = G(T, P, N)} \qquad (21)$$

Therefore, the natural variables of G are T, P, N.

4.2. Themodynamic Potentials

Gibbs free energy

Useful work and minimum condition

The Gibbs free energy will be useful for dealing with systems at constant T and P.[2] First, consider a *closed* composite system in mechanical equilibrium with a *heat source* and a *volume source*; i.e., $T = T_{ext}$ and $P = P_{ext}$). The 'total system' (system + heat source + volume source) is isolated and is in equilibrium. Therefore,

$$0 = dU^{tot} = d(U + U^{HS} + U^{VS}) = dU + T_{ext} dS^{HS} - P_{ext} dV^{VS}$$

$$= dU - T_{ext} dS + P_{ext} dV = dU - TdS + PdV = d(U - TS + PV)$$

$$dG|_{T=T_{ext}, P=P_{ext}} = 0$$

Therefore, **the equilibrium value of any unconstrained internal parameter of a system which is in equilibrium with a heat source and a volume source, minimizes the system's Gibbs free energy at constant T and P equal the sources' temperature and pressure**.

4.2. Themodynamic Potentials

Gibbs free energy

Useful work and minimum condition

Now, let's consider the same system and heat and volume sources, but the system is now connected to a *source of reversible work*. The system's internal energy changes while it does work on the reversible source, while absorbing heat from the heat source and doing work against the volume source, both also in a reversible way:

$$\Delta U = W + Q \text{ with } Q = T\Delta S; \text{ and } W = -W^{useful} + W^{VS} = -W^{useful} - P_{ext}\Delta V$$

wherefrom we get that

$$\Delta U - T\Delta S + P\Delta V = -W^{useful}$$

We see that, **since temperature and pressure are constant**,

$$\boxed{\Delta G = -W^{useful}}$$

That is, **when the system's temperature and pressure are kept constant, the work we can extract from it is equal to the change of the Gibbs free energy**.

4.2. Themodynamic Potentials

Gibbs free energy

Useful work and irreversibility

If the process were irreversible, we can write Clausius's inequality $Q^{irr} < T\Delta S$. But the work on the volume source $-W^{VS}$ is still $-P_{ext}\Delta V = -P\Delta V$

Now using the first principle and this inequality we can write

$$\Delta U - Q^{irr} = W^{irr} - P\Delta V \Rightarrow \Delta G = \Delta U - T\Delta S + P\Delta V < W^{irr}$$

Since the work done by the system is negative, we can conclude that the work we get is less than the change in G, i.e.

$$\boxed{\Delta G < W^{irr}} \text{ o } |\Delta G| > |W^{irr}|$$

Minimum Principle at T and P constant: $\Delta G < W^{irr}$

Then, **in a system at constant T and P, any spontaneous process will produce a decrease of G to its minimum**.

4.2. Themodynamic Potentials

Gibbs free energy

Other properties

If we know G, we can also deduce the state equations. As before, these arise from comparing the differential dG with the equation (21). By comparing term by term we deduce that

$$S = -\left.\frac{\partial G}{\partial T}\right|_{P,N};\ \ V = \left.\frac{\partial G}{\partial P}\right|_{T,N};\ \ \mu = \left.\frac{\partial G}{\partial N}\right|_{T,P} \tag{22}$$

In the same way, we obtain three relations of thermodynamic consistency (Maxwell's relations), which now involve different variables than we obtained with the internal energy:

$$-\left.\frac{\partial S}{\partial P}\right|_{T,N} = \left.\frac{\partial V}{\partial T}\right|_{P,N};\ -\left.\frac{\partial S}{\partial N}\right|_{T,P} = \left.\frac{\partial \mu}{\partial T}\right|_{P,N};\ \ \left.\frac{\partial V}{\partial N}\right|_{T,\cancel{V}} = \left.\frac{\partial \mu}{\partial P}\right|_{T,N}$$

T,P

4.2. Themodynamic Potentials

Relations between potentials

Relations between potentials

The Legendre transformations that lead us to the thermodynamic potentials have all been calculated from the internal energy U. However, from the potentials' definitions, we can find relationships between them that are often useful. Let's discuss some of them:

From the definition of the enthalpy $H \equiv U + PV$ and the Gibbs free energy, $G \equiv U - TS + PV$, we can conclude that

$$G = H - TS$$

We can also consider the Helmholtz free energy $F \equiv U - TS$ and relate it to G, i.e.

$$G = F + PV$$

This type of relationships can be useful in many problems, depending on the information that is accessible to us.

4.3. Third Law

Postulate III

It is a fact of experience that absolute zero temperature is not accessible by any process that has a finite number of stages. This arises from the fact that the variation of entropy, as a function of any variable, in a reversible process near absolute zero, tends to zero. This fact indicates that entropy must be finite (not infinite) at absolute zero. This suggests that an entropy scale can be chosen.

The postulate is fundamentally based on *statistical mechanics*, which relates entropy to disorder. Then, at absolute zero, in a system in its most ordered state, it can only have one state and therefore there will be no disorder, i.e. the entropy is zero.

Postulate III

The entropy of any substance, in its most stable crystalline form, is zero at Absolute Zero temperature.

This fact allows us to fix, therefore, an absolute scale of entropy, which will always be positive on this scale.

4.3. Third Law

Postulate III

Heat capacity's behavior at low temperature

Different studies (Einstein, Debye, etc.) based on statistical mechanics and experimental results of crystalline solids at low temperature, indicated that there was a general pattern in their behavior.

This is because the energy stored in a perfect crystalline solid at low temperature is due to the vibrations of the crystal lattice (in quantum mechanics, these (atom) vibrations are associated with the existence of quasi-particles called *phonons*), for any crystalline solid.

Based on this idea and making calculations from an elastic network model, Peter Debye found that the heat capacity of solids at low temperature is of the form

$$\boxed{C_{Vm} = aT^3} \quad \text{for} \quad T \to 0 \tag{23}$$

where a is a constant that depends on the specific substance [5].

Recommended reading

[1] I. Levine, *Principios de Fisicoquímica*, McGraw-Hill; Chap. 4

[2] H.B. Callen, *Termodinámica*, Ediciones AC; Chaps. 3, 5 and 6.

[3] https://ca.wikipedia.org/wiki/Hermann_von_Helmholtz

[4] https://ca.wikipedia.org/wiki/Josiah_Willard_Gibbs

[5] https://ca.wikipedia.org/wiki/Peter_Debye

[6] https://en.wikipedia.org/wiki/Definite_quadratic_form

[7] F.M. White, *Fluid Mechanics*, McGraw-Hill; Chap. 3